

Nahamcon 2020 Crypto - Twinning

deuterium — 2020-06-14

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/nahamconCTF/crypto/Twinning/2020-06-14-Nahamcon-2020-twinning.html>

Abstract

CTF writeup: A gap of two is poor personal space for RSA primes: a square root factors the modulus in NahamCon 2020's Twinning and unlocks the PIN.

Twining

Challenge

252 Solves

×

Twining

100

These numbers wore the same shirt! LOL, #TWINNING!

Connect with:

nc jh2i.com 50013

Flag

Submit

The name of the challenge is a subtle hint towards Twin primes, i.e. the RSA primes p and q being twin primes, at a difference of two.

How does that help?

We can now easily factor n since we have another relation, namely

$$\begin{aligned} q &= p + 2 \text{ and} \\ p \cdot q &= N \end{aligned}$$

```
q^2 - q*p = N
( q - 1 )^2 = N + 1
q = sqrt( N + 1 ) + 1
p = sqrt( N + 1 ) - 1
```

One can easily solve a quadratic equation to find p

Once p and q are found, rest is a piece of cake

Although the time was not a concern, I still wrote a small script

```
from pwn import remote
import gmpy2
import re

HOST, PORT = "jh2i.com", 50013
REM = remote(HOST, PORT)

data = REM.recvuntil(b'What is the PIN?')
print(data.decode())
e_n = re.search(b'(\d+),(\d+)',data)
e = int(e_n[1])
n = int(e_n[2])
PIN = int(re.search(b'is (\d+)',data)[1])

def factor(n,e):
    a = gmpy2.iroot(n+1,2)[0]
    phi = (a-2)*(a)
    return gmpy2.invert(e,phi) # returns d

DECRYPTED_PIN = pow(PIN, factor(n,e), n)
REM.sendline(str(DECRYPTED_PIN).encode())
print(REM.recvline().decode())
print(REM.recvline().decode())
print(REM.recvline().decode())
```

flag—thats the twinning pin to win”

Interactive controls are omitted from this PDF. See the canonical HTML version.